

Lecture 6: Hamilton-Jacobi-Bellman Equations

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6.1 The Dynamic Programming Principle

Let us define the optimal value function,

$$H(t, x) = \sup_{u \in \mathcal{A}_{t,T}} H^u(t, x) , \quad (6.1)$$

where the objective is $H^u(t, x) := \mathbb{E}_{t,x} \left[G(X_T^u) + \int_t^T F(s, X_s^u, u_s) ds \right]$, subject to the state dynamics

$$dX_t^u = \mu(t, X_t^u, u_t) dt + \sigma(t, X_t^u, u_t) dW_t , \quad (6.2)$$

where:

- u_t is the agent's control process,
- X_t^u is the controlled state process,
- W_t is a standard 1D Brownian motion,
- G is the terminal reward function,
- F is the running reward/penalty rate,
- $\mathcal{A}_{t,T}$ is the set of admissible predictable strategies yielding a unique strong solution to (6.2),
- $\mathbb{E}_{t,x}[\cdot]$ denotes expected value conditional on $X_t^u = x$.

We recall the **Dynamic Programming Principle (DPP)** over any stopping time $\tau \in [t, T]$:

$$H(t, x) = \sup_{u \in \mathcal{A}_{t,T}} \mathbb{E}_{t,x} \left[H(\tau, X_\tau^u) + \int_t^\tau F(s, X_s^u, u_s) ds \right] . \quad (6.3)$$

6.2 Derivation of the Hamilton-Jacobi-Bellman (HJB) Equation

To derive the local differential condition from the global DPP, let us choose small parameters $h > 0$ and $\epsilon > 0$. Given $X_t^u = x$ and any arbitrary control $u \in \mathcal{A}_{t,T}$, define the stopping time:

$$\tau = T \wedge \inf \{s > t \mid (s - t, |X_s^u - x|) \notin [0, h) \times [0, \epsilon)\} .$$

Assuming $H(t, x) \in C^{1,2}([0, T] \times \mathbb{R})$, applying Itô's Lemma to $H(\tau, X_\tau^u)$ yields:

$$H(\tau, X_\tau^u) = H(t, x) + \int_t^\tau \left(\frac{\partial}{\partial t} + \mathcal{L}_s^u \right) H(s, X_s^u) ds + \int_t^\tau \sigma(s, X_s^u, u_s) \frac{\partial}{\partial x} H(s, X_s^u) dW_s , \quad (6.4)$$

where $\mathcal{L}_t^u$ is the infinitesimal generator operator:

$$\mathcal{L}_t^u := \mu(t, x, u_t) \frac{\partial}{\partial x} + \frac{1}{2} \sigma^2(t, x, u_t) \frac{\partial^2}{\partial x^2} .$$

From the lower bound of the DPP, any choice of control u satisfies $H(t, x) \geq \mathbb{E}_{t,x} [H(\tau, X_\tau^u) + \int_t^\tau F(s, X_s^u, u_s) ds]$. Substituting (6.4) into this lower bound and assuming the stochastic integral is a martingale (hence having zero expectation):

$$H(t, x) \geq \mathbb{E}_{t,x} \left[H(t, x) + \int_t^\tau \left(\frac{\partial}{\partial t} + \mathcal{L}_s^u \right) H(s, X_s^u) ds + \int_t^\tau F(s, X_s^u, u_s) ds \right] .$$

Subtracting $H(t, x)$ from both sides, dividing by h , and taking the limit as $h \searrow 0$ (applying the Mean Value Theorem / Mean Rate of Change inside the expectation):

$$\begin{aligned} 0 &\geq \lim_{h \searrow 0} \frac{1}{h} \mathbb{E}_{t,x} \left[\int_t^\tau \left(\left(\frac{\partial}{\partial t} + \mathcal{L}_s^u \right) H(s, X_s^u) + F(s, X_s^u, u_s) \right) ds \right] \\ &= \left(\frac{\partial}{\partial t} + \mathcal{L}_t^u \right) H(t, x) + F(t, x, u_t) , \end{aligned}$$

with equality holding if u_t is an optimal control u^* .

Taking the supremum over all admissible instant actions $u \in \mathcal{A}_{t,x}$, we obtain the **Hamilton-Jacobi-Bellman (HJB) Equation**:

$$\frac{\partial H}{\partial t}(t, x) + \sup_{u \in \mathcal{A}_{t,x}} (\mathcal{L}_t^u H(t, x) + F(t, x, u)) = 0 , \quad (6.5)$$

subject to the terminal condition $H(T, x) = G(x)$, where $\mathcal{A}_{t,x}$ denotes the set of feasible control decisions at (t, x) .

6.3 Example: Merton's Optimal Investment Problem

Consider an investor seeking to maximize terminal utility of wealth. The asset price S_t and portfolio wealth X_t^π follow

$$\begin{aligned} dS_t &= \mu S_t dt + \sigma S_t dW_t \\ dX_t^\pi &= r X_t^\pi dt + (\mu - r) \pi_t dt + \sigma \pi_t dW_t , \end{aligned}$$

where π_t is the dollar amount invested in the risky asset.

The value function parameterized by time, wealth, and stock price is:

$$H(t, x, s) = \sup_{\pi \in \mathcal{A}_{t,T}} \mathbb{E}_{t,x,s} [U(X_T^\pi)] .$$

Applying the 2D Itô Lemma (taking into account that X_t^π and S_t share the exact same Brownian motion dW_t , yielding a cross-volatility term $\sigma \pi_t \cdot \sigma s = \sigma^2 \pi_t s$):

$$\left(\frac{\partial}{\partial t} + rx \frac{\partial}{\partial x} + \mu s \frac{\partial}{\partial s} + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2}{\partial s^2} \right) H + \sup_{\pi} \left((\mu - r) \pi \frac{\partial H}{\partial x} + \frac{1}{2} \sigma^2 \pi^2 \frac{\partial^2 H}{\partial x^2} + \sigma^2 \pi s \frac{\partial^2 H}{\partial x \partial s} \right) = 0 , \quad (6.6)$$

with terminal condition $H(T, x, s) = U(x)$.

Step 1: Dimensional Reduction

Because neither the wealth dynamics dX_t^π nor the terminal condition $U(x)$ depend on the stock price S_t , we propose an ansatz independent of s : $H(t, x, s) = h(t, x)$.

Substituting $h(t, x)$ eliminates all partial derivatives with respect to s , reducing (6.6) to:

$$\frac{\partial h}{\partial t} + rx \frac{\partial h}{\partial x} + \sup_{\pi} \left((\mu - r)\pi \frac{\partial h}{\partial x} + \frac{1}{2} \sigma^2 \pi^2 \frac{\partial^2 h}{\partial x^2} \right) = 0, \quad h(T, x) = U(x). \quad (6.7)$$

Step 2: First-Order Condition for Optimal Portfolio

Assuming strictly concave value function $h_{xx}(t, x) < 0$, the expression inside the supremum is quadratic in π . Differentiating with respect to π and setting to zero yields the optimal feedback control:

$$\pi^* = -\frac{(\mu - r) \frac{\partial h}{\partial x}}{\sigma^2 \frac{\partial^2 h}{\partial x^2}}. \quad (6.8)$$

Substituting π^* back into (6.7) yields the non-linear PDE:

$$\frac{\partial h}{\partial t} + rx \frac{\partial h}{\partial x} - \frac{(\mu - r)^2 \left(\frac{\partial h}{\partial x} \right)^2}{2\sigma^2 \frac{\partial^2 h}{\partial x^2}} = 0. \quad (6.9)$$

Step 3: Explicit Solution for Exponential (CARA) Utility

Consider CARA utility $U(x) = -e^{-\gamma x}$ for risk aversion parameter $\gamma > 0$. We test the separable ansatz:

$$h(t, x) = -\alpha(t)e^{-\gamma x \beta(t)}, \quad \alpha(T) = 1, \quad \beta(T) = 1.$$

Computing the partial derivatives:

$$\begin{aligned} \frac{\partial h}{\partial t} &= -\alpha'(t)e^{-\gamma x \beta(t)} + \alpha(t)\gamma x \beta'(t)e^{-\gamma x \beta(t)}, \\ \frac{\partial h}{\partial x} &= \alpha(t)\gamma \beta(t)e^{-\gamma x \beta(t)}, \quad \frac{\partial^2 h}{\partial x^2} = -\alpha(t)\gamma^2 \beta^2(t)e^{-\gamma x \beta(t)}. \end{aligned}$$

Substituting these into (6.9) and dividing by $-h(t, x) = \alpha(t)e^{-\gamma x \beta(t)}$:

$$\frac{\alpha'(t)}{\alpha(t)} - \gamma x \left(\beta'(t) + r\beta(t) \right) - \frac{(\mu - r)^2}{2\sigma^2} = 0.$$

For this equation to hold for all x , the coefficient of x and the remaining terms must independently vanish:

$$\beta'(t) + r\beta(t) = 0 \implies \beta(t) = e^{r(T-t)}, \quad (\text{since } \beta(T) = 1) \quad (6.10)$$

$$\alpha'(t) - \frac{(\mu - r)^2}{2\sigma^2} \alpha(t) = 0 \implies \alpha(t) = e^{-\frac{(\mu - r)^2}{2\sigma^2}(T-t)}, \quad (\text{since } \alpha(T) = 1). \quad (6.11)$$

Plugging h_x and h_{xx} into the optimal policy formula (6.8), we find the optimal risky dollar allocation:

$$\pi_t^* = \frac{\mu - r}{\gamma \sigma^2 \beta(t)} = \frac{\mu - r}{\gamma \sigma^2} e^{-r(T-t)}.$$

6.4 Remarks

1. **Feedback Controls:** The optimal strategy π_t^* is obtained in *feedback form*—expressed as a function of the value function derivatives. For CARA utility, the optimal dollar allocation π_t^* is independent of current wealth X_t , reflecting constant absolute risk aversion.
2. **The Hamiltonian:** The optimized term inside the HJB equation,

$$\mathcal{H}(t, x, p, M) = \sup_{u \in \mathcal{A}_{t,x}} \left(\mu(t, x, u)p + \frac{1}{2} \sigma^2(t, x, u)M + F(t, x, u) \right),$$

is called the **Hamiltonian**. Its arguments are time t , position x , gradient $p = DH$, and Hessian matrix $M = D^2H$.

3. **Utility Classes:** Exponential utility exhibits Constant Absolute Risk Aversion (CARA), where the Arrow-Pratt measure $-U''(x)/U'(x) = \gamma$ is constant. Other common utility classes include Constant Relative Risk Aversion (CRRA, e.g., $U(x) = \frac{x^{1-\gamma}}{1-\gamma}$ with $-xU''(x)/U'(x) = \gamma$) and Hyperbolic Absolute Risk Aversion (HARA).
4. **References:** This lecture follows Chapter 5.3 of [1]; for rigorous stochastic control foundations, see [2].

References

- [1] Á. Cartea, S. Jaimungal, and J. Penalva. *Algorithmic and high-frequency trading*. Cambridge University Press, 2015.
- [2] W. H. Fleming and H. M. Soner. *Controlled Markov processes and viscosity solutions*. Springer, 2006.